

JOINS

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22.9.26

DBMS LAB 5-SQL JOINS

DATABASE:

```
1 • CREATE DATABASE JOINSSS;
2 • USE JOINSSS;
3 • CREATE TABLE Departments (
4     department_id INT PRIMARY KEY,
5     department_name VARCHAR(50)
6 );
7
8
9 • CREATE TABLE Employees (
10     employee_id INT PRIMARY KEY,
11     employee_name VARCHAR(50),
12     department_id INT,
13     manager_id INT
14 );
15 • SHOW TABLES;
16
17
18 • INSERT INTO Departments (department_id, department_name) VALUES
19     (10, 'Engineering'),
20     (20, 'Marketing'),
21     (30, 'Sales'),
22     (40, 'Human Resources');
23
24 INSERT INTO Employees (employee_id, employee_name, department_id, manager_id) VALUES
25     (1, 'Alice', 10, NULL),
26     (2, 'Bob', 10, 1),
27     (3, 'Charlie', 20, 1),
28     (4, 'David', NULL, 2);
29 SELECT * FROM Departments;
30 SELECT * FROM Employees;
```

	department_id	department_name
▶	10	Engineering
	20	Marketing
	30	Sales
	40	Human Resources
*	NULL	NULL

Result Grid					Filter Rows:	Edit:
	employee_id	employee_name	department_id	manager_id		
▶	1	Alice	10	NULL		
	2	Bob	10	1		
	3	Charlie	20	1		
	4	David	NULL	2		
*	NULL	NULL	NULL	NULL		

1) INNER JOIN:

```

1 • SELECT * FROM Departments;
2 • SELECT * FROM Employees;
3 • SELECT
4     e.employee_id,
5     e.employee_name,
6     d.department_name
7 FROM Employees e
8 INNER JOIN Departments d
9     ON e.department_id = d.department_id;

```

	employee_id	employee_name	department_name
▶	1	Alice	Engineering
	2	Bob	Engineering
	3	Charlie	Marketing

2) LEFT JOIN:

```

SELECT
    e.employee_id,
    e.employee_name,
    d.department_name
FROM Employees e
LEFT JOIN Departments d
    ON e.department_id = d.department_id;

```

	employee_id	employee_name	department_name
▶	1	Alice	Engineering
	2	Bob	Engineering
	3	Charlie	Marketing
	4	David	NULL

3)RIGHT JOIN:

```

1 • SELECT
2     e.employee_id,
3     e.employee_name,
4     d.department_name
5 FROM Employees e
6 RIGHT JOIN Departments d
7     ON e.department_id = d.department_id;

```

	employee_id	employee_name	department_name
▶	2	Bob	Engineering
	1	Alice	Engineering
	3	Charlie	Marketing
	NULL	NULL	Sales
	NULL	NULL	Human Resources

4)FULL OUTER JOIN:

```

SELECT e.employee_id, e.employee_name, d.department_name
FROM Employees e
LEFT JOIN Departments d ON e.department_id = d.department_id

UNION

SELECT e.employee_id, e.employee_name, d.department_name
FROM Employees e
RIGHT JOIN Departments d ON e.department_id = d.department_id;

```

	employee_id	employee_name	department_name
▶	1	Alice	Engineering
	2	Bob	Engineering
	3	Charlie	Marketing
	4	David	NULL
	NULL	NULL	Sales
	NULL	NULL	Human Resources

5)SELF JOIN:

- **SELECT**
emp.employee_id,
emp.employee_name,
mgr.employee_name
FROM Employees emp
JOIN Employees mgr
ON emp.manager_id = mgr.employee_id;

Result Grid	Filter Rows:	Exp
employee_id	employee_name	employee_name
2	Bob	Alice
3	Charlie	Alice
4	David	Bob

6)CROSS JOIN:

- 1 • **SELECT**
2 e.employee_name,
3 d.department_name
4 **FROM** Employees e
5 **CROSS JOIN** Departments d;

Result Grid	Filter Rows:
employee_name	department_name
David	Engineering
Charlie	Engineering
Bob	Engineering
Alice	Engineering
David	Marketing
Charlie	Marketing
Bob	Marketing
Alice	Marketing
David	Sales
Charlie	Sales
Bob	Sales
Alice	Sales
David	Human Resources
Charlie	Human Resources
Bob	Human Resources
Alice	Human Resources

QUIRES AND SUBQUIRES

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DBMS LAB 6-QUIRES AND SUBQUIRES:

```
• CREATE DATABASE SchoolDB;
• USE SchoolDB;

• CREATE TABLE Student (
    ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Marks INT,
    Department VARCHAR(50)
);
• SHOW TABLES;
• INSERT INTO Student (ID, Name, Marks, Department) VALUES
    (1, 'Amit', 85, 'CS'),
    (2, 'Shrujana', 92, 'CS'),
    (3, 'Surabhi', 97, 'IT'),
    (4, 'Aditya', 65, 'EC'),
    (5, 'Sanjana', 78, 'IT');

• SELECT * FROM Student;

• SELECT Name, Marks
    FROM Student
    WHERE Marks > 80;

• SELECT Name, Marks
    FROM Student
    WHERE Marks > (SELECT AVG(Marks) FROM Student);
```



```
SELECT Name, Marks
FROM Student
WHERE Marks = (SELECT MAX(Marks) FROM Student);
```

```
SELECT Name, Department
FROM Student
WHERE Department = (SELECT Department FROM Student WHERE Name = 'Shrujana');
```

```
SELECT Name, Department
FROM Student
WHERE Department IN (SELECT Department FROM Student WHERE Marks > 80);
```

```
SELECT Name, Marks
FROM Student
WHERE Marks > ALL (SELECT Marks FROM Student WHERE Department = 'EC');
```

Outputs-

1. Basic query-

Result Grid				
Filter Rows:				
	ID	Name	Marks	Department
▶	1	Amit	85	CS
	2	Shrujana	92	CS
	3	Surabhi	97	IT
	4	Aditya	65	EC
	5	Sanjana	78	IT
*	NULL	NULL	NULL	NULL

2. Conditional query-

Result Grid		
Filter R		
	Name	Marks
▶	Amit	85
	Shrujana	92
	Surabhi	97

3. Single-value subquery -

Result Grid		
Filter		
	Name	Marks
▶	Amit	85
	Shrujana	92
	Surabhi	97

4. Aggregate subquery -

Result Grid   F

	Name	Marks
▶	Surabhi	97

5. Related-value subquery -

Result Grid   Filter ROW

	Name	Department
▶	Amit	CS
	Shrujana	CS

6. Multi-row subquery -

Result Grid   Filter ROW

	Name	Department
▶	Amit	CS
	Shrujana	CS
	Surabhi	IT
	Sanjana	IT

7. Multi-row comparison subquery -

Result Grid   Filter

	Name	Marks
▶	Amit	85
	Shrujana	92
	Surabhi	97
	Sanjana	78
